Discrete Maps and Chaos: Exercise Guide for 4A

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Mathematical Methods for Physicists II

Python Notebook Guide

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This is a collection of notes accompanying the discrete-map exercises from my Mathematical Methods for Physicists II coursework. It follows notebook 4A and explains the reasoning behind each algorithm, plot, and interpretation, including the revised bifurcation starting value. The companion theoretical foundations introduce the necessary mathematical concepts first.

1 How to use this exercise guide

Read *Discrete Maps and Chaos: Conceptual Foundations for 4A* for invariant intervals, fixed points, multipliers, periodic orbits, cobwebs, and sensitivity. This guide follows the revised notebook's five exercise sections and explains how to reconstruct each calculation.

No random sampling appears in these exercises. Variability is produced by deterministic iteration. The notebook's parameter cutoffs protect its displayed runs; they are not physical boundaries. The revised bifurcation scan also avoids a special fixed-point initial state, explained in the final exercise below.

2 Exercise 1: logistic population sequences

2.1 Specify the experiment

Use $x_{n+1} = \mu x_n(1 - x_n)$, $x_0 = 0.75$, and 30 updates for each $\mu = 2.8, 3.3, 3.5, 3.6, 3.8$. All preserve the unit interval. Store the initial state in a list, then repeatedly use the latest list entry to calculate and append the next state.

In Python `values[-1]` selects the current last entry. A loop of 30 updates produces 31 values, indexed from zero through 30. Plot them against these iteration indices, rather than pretending the values are continuous-time samples of an integrated differential equation.

2.2 Interpret the panels

For $\mu = 2.8$, the nonzero fixed point is about 0.6429 with multiplier -0.8, so late deviations shrink while alternating sides. For 3.3 and 3.5, stable period-two and period-four behavior is expected after transients. The 3.6 and 3.8 examples can display irregular late behavior, but 30 updates offer only a short view.

A connected line joins discrete samples for readability; it does not define the model between updates. A short alternating trace can still be transient, and a complicated trace alone does not measure chaos. Longer histories and the later bifurcation plot give additional context.

3 Exercise 2: implement and read cobwebs

3.1 Trace a forward update

The common function builds lists of horizontal and vertical coordinates for a staircase. Initially it stores $(x_0, 0)$. For an ordinary forward cobweb:

1. Evaluate $y = f(x)$ and append (x, y), the vertical move to the curve.
2. Replace x by y and append (x, y) again, now on the diagonal.
3. Repeat with that new horizontal coordinate.

The lists are coordinates of one drawing, not independent x and y solutions. Each update appends two points, so N steps give $2N + 1$ drawing coordinates.

3.2 Squaring, the square root, and inverse steps

The first comparison begins at $x_0 = 0.0001$ and takes ten steps. Against the curve $y = x^2$, the function's inverse option first moves vertically to the diagonal and then horizontally to the curve using $x = \sqrt{y}$. Its horizontal states therefore follow square-root iteration, increasing toward one.

The second panel draws $y = \sqrt{x}$ explicitly and uses ordinary forward steps. It gives the same sequence but a different geometric staircase. Neither panel shows forward squaring approaching one. Forward squaring of a state strictly between zero and one would decrease toward zero.

The inverse option needs an appropriate inverse branch; the square root is unique on the nonnegative domain being used. The function does not automatically solve arbitrary inverse problems or resolve multiple branches

for another map.

3.3 Logistic cobwebs

Define a callable for $f(x) = \mu x(1 - x)$ at the current parameter. Use $x_0 = 0.2$ and 30 updates at 2.8 and 3.8. Draw the function, diagonal, and forward staircase.

At 2.8 the staircase contracts toward the fixed-point intersection. At 3.8 it does not settle onto the repelling nonzero fixed point. The negative fixed-point slopes explain alternating departures or approaches. These pictures visualize the same iteration rule as a time sequence, not a different dynamical model.

4 Exercise 3: compare four other maps

4.1 Build a common dispatcher

A function takes a map name, parameter, initial state, and maximum update count. Initialize a values list. On each pass read the last state, choose the formula associated with the name, calculate the new state, append it, and then stop if it is nonfinite or has magnitude above 100. An unrecognized name raises an error rather than returning an unrelated rule.

Append before checking the threshold so the first escaping state remains visible. Do not interpret the final saved value as exactly 100 or an accurately located crossing. Also note that arbitrary parameter changes could overflow inside a formula before this post-update safeguard runs.

4.2 Use the actual rules and durations

All begin at 0.75 and use $\mu = 2.8, 3.3, 3.5, 3.8$, but the formulas and update counts differ:

Map	Rule	Maximum updates
Ecology/Ricker	$xe^{\mu(1-x)}$	30
Gaussian	$e^{-3x^2} + \mu$	15
Tent	$\mu[1 - 2\lvert x - 1/2\rvert]$	10
Quartic	$\mu[1 - (2x - 1)^4]$	5

The Ricker map stays positive and can show strong growth followed by suppression. Its positive equilibrium at one is not linearly attracting for these parameters, since its multiplier is $1 - \mu$.

The Gaussian parameter shifts the entire output upward. All outputs lie in $(\mu, \mu + 1]$, and the displayed large positive shifts make the exponential correction tiny after a few updates. Rapid settling near μ is not surprising, even when the same number gives irregularity in a logistic map.

The tent and quartic examples leave the intended unit interval because these parameters exceed their interval-preserving range. Their escape and cutoff messages must not be called bounded chaos. The maximum durations are different, so these panels are not a controlled comparison of rates of convergence across all four maps.

4.3 Repeat the tent and quartic cases in an invariant range

Use $\mu = 0.6, 0.9, 1.0$, $x_0 = 0.75$, and 40 updates for each tent or quartic rule. Now $0 \le \mu \le 1$ ensures interval preservation, so the plots address bounded dynamics.

The tent $\mu = 1$ curve has a special exact sequence: $0.75, 0.5, 1, 0, 0, \ldots$. Its collapse does not prove global attraction to zero. A dyadic start and finite binary precision are essential to interpreting this output. The quartic map has a different derivative and need not share this special trajectory.

The revised explanatory text states the complete nonnegative parameter interval, rather than the incomplete condition $\mu \le 1$, which would also admit negative parameters.

5 Exercise 4: fixed points and nearby starts

5.1 What changes between the two trajectories

For each $\mu = 2.8, 3.8$, calculate 80 logistic updates from 0.2 and from 0.200001. The parameter and iteration count are identical; only the initial state changes. At each common index calculate $d_n = |x_n - y_n|$.

Plot d_n on a logarithmic vertical scale to reveal early small differences. In the revised code a zero difference is mapped to NaN for display and omitted. This avoids taking a logarithm of zero without pretending the measurement was a chosen positive floor.

5.2 What to conclude

At 2.8 both orbits enter the same attracting fixed point's basin and their separation decreases after transients. At 3.8 the initially small difference can grow strongly and then fluctuate on the scale of the interval.

The fixed-point multiplier $2 - \mu$ describes perturbations near that fixed point. It is not a universal per-step growth factor along a wandering chaotic orbit. The foundations' product of trajectory-dependent derivatives gives that distinction.

A roughly straight rising segment on a semilog plot is compatible with exponential separation over that segment. Later saturation and fluctuations cannot be fitted by one exponential growth law. This finite experiment illustrates sensitivity, not a numerical estimate of a Lyapunov exponent or a proof for every initial condition.

6 Exercise 5: reconstruct the bifurcation scan

6.1 Separate the parameter loop and iteration loop

Create 220 evenly spaced values from 2.6 to 4.0. For each parameter reset the initial state to 0.751, then take 450 updates. Save a pair (μ, x) only when the zero-based loop counter is at least 350. The retained states are therefore x_{351} through x_{450}, not the initial state or all 450 points.

The two stored arrays have matching lengths: 100 pairs for each of 220 parameters, hence 22,000 plotted pairs. Resetting the start at each parameter keeps this experiment distinct from a continuation method that initializes each parameter from the preceding run's final state.

6.2 Why the starting value was corrected

The original scan used 0.75. At exactly $\mu = 4$ it is an unstable fixed point, and it is exactly representable in binary. The code would remain there and display a single endpoint level unrelated to generic late-time behavior at that parameter.

Using 0.751 avoids that particular exceptional orbit while leaving the model, parameter grid, and discard/retention counts unchanged. It is used only for this scan; the other exercises retain their stated starting values. The new value does not make finite-precision iteration exact or guarantee that every parameter has been fully explored.

6.3 Read the geometry of the result

A stable fixed-point branch gives one late level, a stable period-two branch two levels, and repeated period doublings more branches. Chaotic bands can contain periodic windows. Darker plotted regions can result from repeated visits and overlapping markers, but are not calibrated probabilities.

Near a threshold the fixed 350-update discard may leave slow transients. A 220-point parameter grid can skip small windows entirely. If a narrow feature matters, increase transient lengths and parameter resolution locally and inspect individual trajectories. The present diagram is an introductory survey.

7 Checks before interpreting your own maps

Verify the state domain and parameter normalization for each formula. Check the number of stored states, the distinction between forward and inverse iteration, and any known analytic fixed points. Test whether a suspicious simple sequence is an exceptional exact orbit before claiming global stability.

Separate numerical safeguards, finite-precision artifacts, and mathematical escape from genuine bounded dynamics. Use longer late-time histories to examine apparent cycles, and avoid reading a normalized invariant distribution or a quantitative chaos measure from marker density alone.